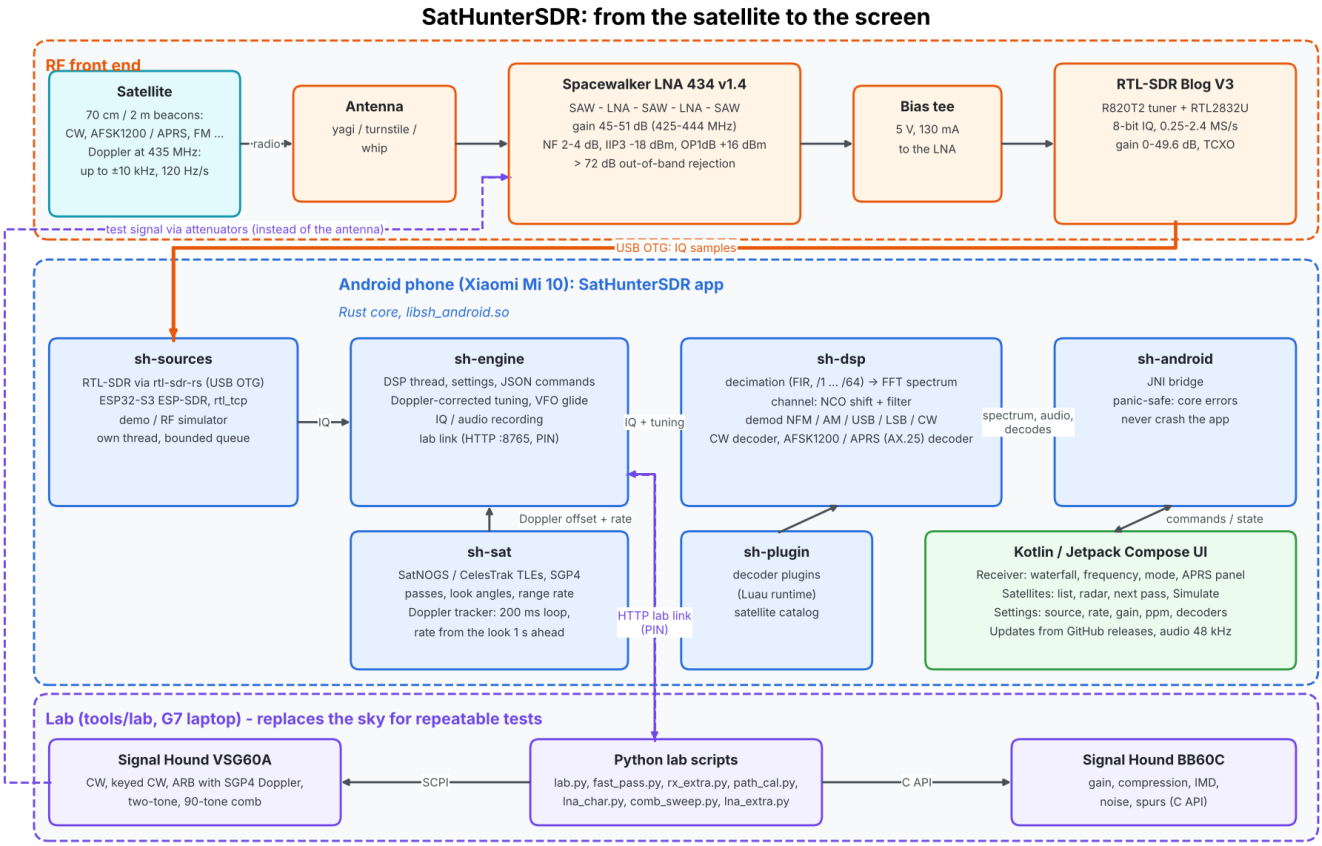
SatHunterSDR + Spacewalker LNA 434: what the lab and the field showed

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SatHunterSDR is an Android app that turns a phone, an RTL-SDR and the Spacewalker LNA 434 (Decent Labs) into a CubeSat receiver: it picks the satellite, predicts the pass with SGP4, follows the Doppler and decodes CW and APRS on the phone. Over five days we heard satellites in the field, then rebuilt the sky in the lab (Signal Hound VSG60A playing whole passes with their Doppler, BB60C measuring the LNA alone), fixed what the measurements exposed, and measured again.



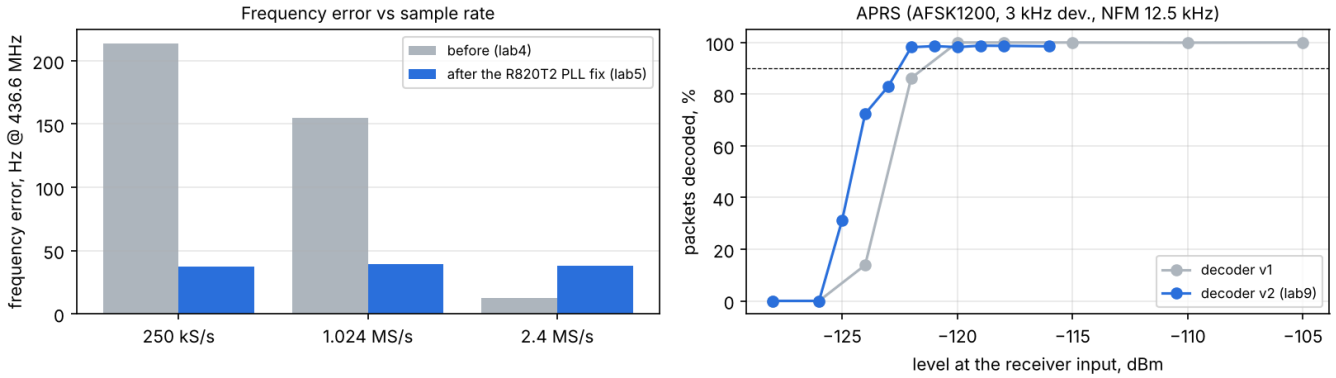
System block diagram: RF chain, Rust core, Android UI and the lab instruments that replace the sky.

Key results

	Quantity	Result
Field	ITUPSAT 1, 70 cm CW, 19° pass, 6-el. yagi	callsign decoded at 12 WPM, C/N0 ~41 dB-Hz, Doppler tracked
Field	CUBEBUG-2, AFSK1200, IQ recording	3 of 5 frames with good CRC, C/N0 ~55 dB-Hz; burst kept as a test vector
Lab	System noise figure (LNA + RTL-SDR, 436.6 MHz)	2.0 dB
Lab	Sensitivity / CW decode	MDS -154.4 dBm in 31 Hz; 15 WPM CW decoded down to -145 dBm (29 dB-Hz)
Lab	APRS (AFSK1200)	90 % packets at -122.5 dBm (3 kHz dev.), -123.2 dBm (1 kHz dev., 6 kHz channel)
Lab	Simulated 79° LEO pass, 116 Hz/s Doppler	receiver within ±9 Hz (1σ) of the satellite; CW decoded through the closest point at -130 dBm
Lab	Orbit model check	phone vs independent SGP4: 0.16 Hz apart at 435 MHz
Lab	Frequency accuracy / decimation	0.09 ppm at every sample rate; alias rejection > 48 dB in the central 84 % of the span
Lab	Spacewalker LNA 434 v1.4 (alone, 5.0 V, 131 mA)	gain 45-51 dB in 425-444 MHz, NF 2-4 dB (two methods, see p. 3), OP1dB +15.5...+21 dBm, IIP3 about -18 dBm, > 72 dB out-of-band rejection, gain stable 0.15 dB over 10 min, no oscillation

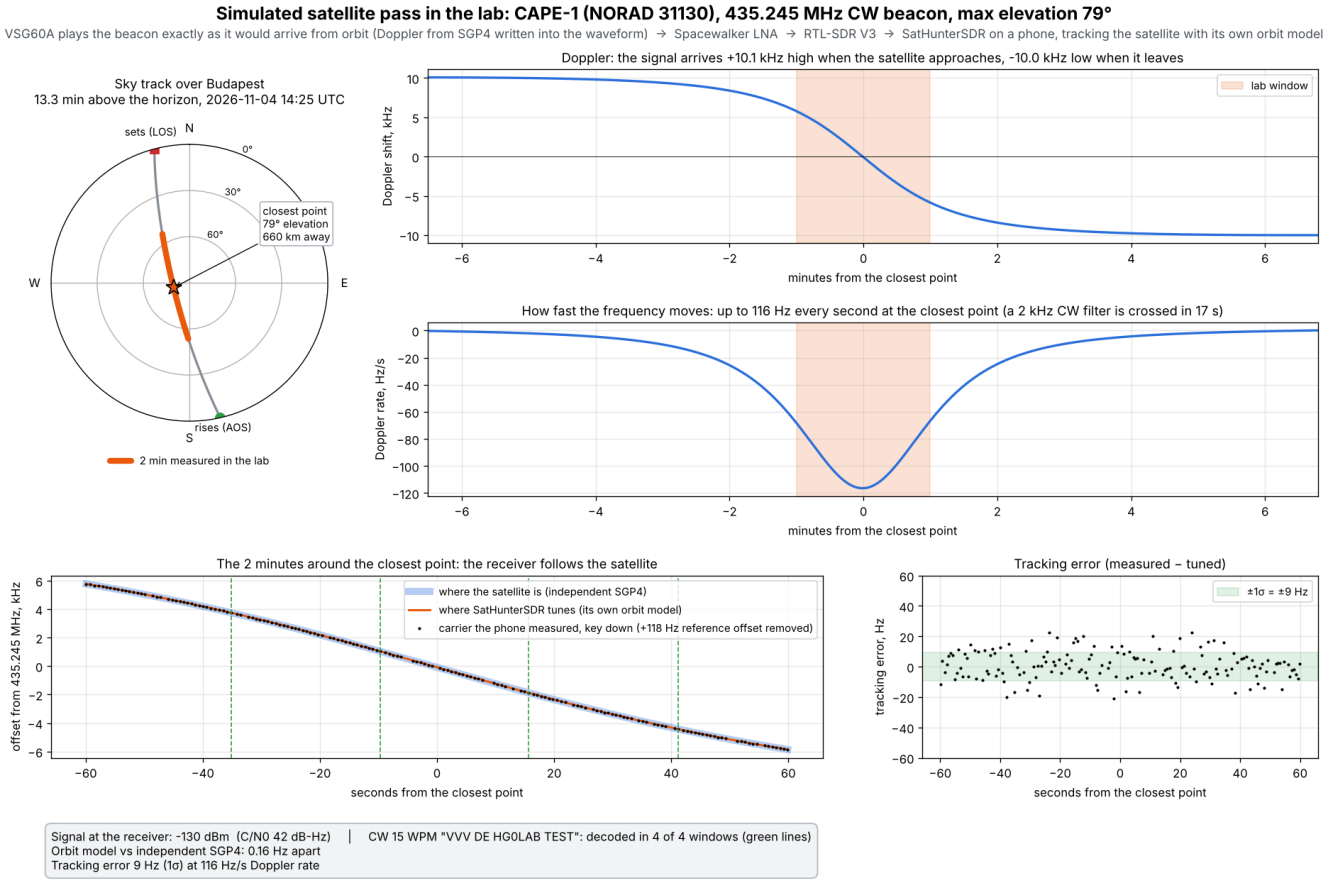
Bugs and weak spots found, and what was done

Found	How it showed up	Fix	Result
Frequency error depended on the sample rate	VSG60 CW at every rate: +10 / +155 / +210 Hz	R820T2 fractional-N (SDM) residual computed in Hz and applied as an IF correction in rtl-sdr-rs	+40 Hz (0.09 ppm) at every rate
Weak beacons hard to see	Field: ITUPSAT 1 barely visible	Spectrum averages every FFT between frames (one FFT per frame skipped ~97 % of the samples)	a beacon 10 dB weaker is visible
Choppy audio, jumpy waterfall	Field: CW demodulator took a whole CPU core, blocks dropped silently	CW 10x, SSB / FM 3x lighter; dropped blocks reported	no silent drops
AM audio capped at 4 kHz; NFM volume depended on the filter	DSP check, modes x bandwidths	AM audio cut follows the bandwidth (1-15 kHz); NFM full scale fixed at 5 kHz deviation	flat response, constant level
Doppler applied in 200 ms steps	Doppler ramps: up to 23 Hz tone jumps at 116 Hz/s	VFO glides at the rate predicted 1 s ahead	±9 Hz (1σ) through a 79° pass
APRS limited by the FM-discriminator detector	APRS sweeps: Eb/N0 ~19 dB at 90 %	650 Hz post-detection filter, DPLL, soft repair of the 24 least confident bits	+1.0 ... +1.5 dB
Build crashed at start	lab7: engine pulled plugin code needing libc++_shared.so	APRS position code moved to sh-dsp (the plugin runtime must package libc++)	fixed in lab8
Retune lost while the RTL-SDR was opening	Field: right frequency shown, wrong one received	retune queued until the device is open	fixed (beta 9)
Lab: generator could not follow a LEO pass	VSG60 retune over SCPI interrupts the output (up to 13.5 s); 20 kS/s ARB plays 16 % slow	whole pass Doppler written into one ARB at 25 kS/s, started on the phone's simulated clock	valid fast-pass test
LNA: 0.22 V drop in the bias feed	supply sweep: 5.22 V needed at the supply for 5.0 V on the board	documented; compensate on long or thin feeds	only 0.2 dB gain at 4.8 V
LNA: 60 dB dropouts, rare	1 in 315 readings, often at high drive; 0 in 5027 with the LNA out	open: LNA, supply or bias-tee DC path	readings re-checked
LNA noise figure: two methods disagree	BB60C gain method 2.6-4.1 dB vs 2.0 dB for the whole chain (phone)	both absolute references checked (input path measured, 50 Ω load = generator off); a calibrated noise source (Y-factor) is next	quoted as 2-4 dB
LNA: 5.5 dB passband ripple, dip at 436 MHz	comb sweep in 200 kHz steps	property of the SAW - LNA - SAW - LNA - SAW cascade; OP1dB is lowest (+15.5 dBm) at the dip	feedback to the design



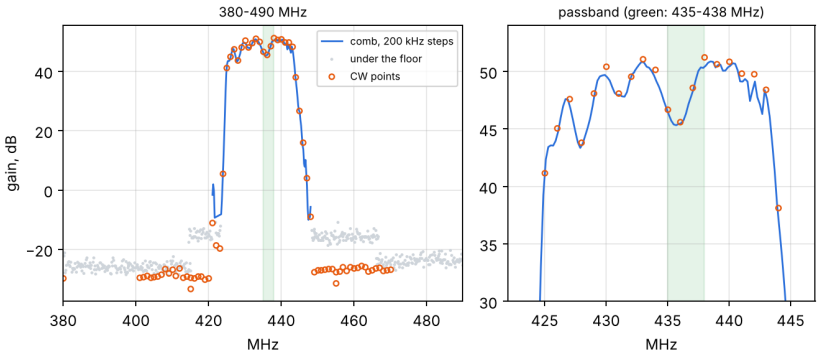
Left: frequency error before and after the R820T2 PLL fix (the remaining +40 Hz is the RTL / generator reference difference, the same at every rate). Right: APRS packet success, decoder v1 and v2.

A satellite pass, replayed in the lab



CAPE-1, 79° pass over Budapest. The VSG60A plays the beacon with the Doppler computed by an independent SGP4; the phone tracks with its own orbit model. Tracking error ±9 Hz (1σ) at 116 Hz/s; CW decoded in all four windows.

Spacewalker LNA 434 v1.4 on its own



Parameter	Result
Gain 435-438 MHz	45.3-50.4 dB
Passband (-10 dB)	425.1-443.7 MHz
Out-of-band rejection	72 to > 80 dB
Noise figure	2.6-4.1 dB (BB60C gain method, 50 Ω load); 2.0 dB from the chain; Y-factor next
IP1dB / OP1dB	-27...-32 / +15.5...+21 dBm
IIP3 / OIP3, 436 MHz	about -18 / +27 dBm
Blocker ±15-18 MHz, -18 dBm	< 0.25 dB desense
Harmonics (2f, 3f)	< -64 dBc
4.5 / 5.0 / 5.5 V on the board	117 / 131 / 146 mA

Transfer function from a 90-tone comb (380-490 MHz, 200 kHz steps) with the CW points; levels at the LNA's connectors, path calibrated without the LNA.

Next

Track down the LNA dropouts (supply, bias tee, LNA); RTL-SDR warm-up drift from a cold start; a coherent AFSK detector for a few more dB on APRS; S11 / S22 with a VNA; more field passes with the campaign hardware.